

DodgeUI: An Adaptive Interface for Mitigating Attentional Conflict via Implicit Motion Cues in Mobile AR

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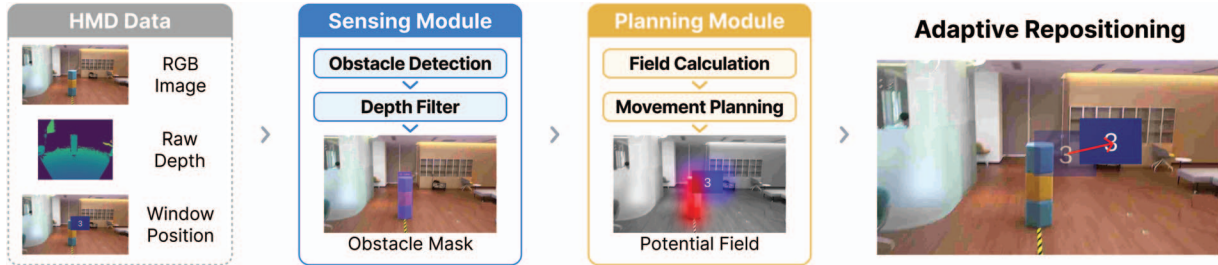


Figure 1: DodgeUI System Pipeline.

ABSTRACT

Interacting with Mobile Augmented Reality (AR) during locomotion creates a fundamental attentional conflict with physical obstacles. We introduce DodgeUI, a context-aware interface driven by Artificial Potential Fields (APF) that adaptively dodges physical hazards. Our system leverages the window's movement as an implicit environmental cue without adding visual elements. A user study (N=24) shows that although objective performance remains stable, participants significantly prefer this adaptive behavior in high-risk scenarios. This work demonstrates how intrinsic attributes of motion can serve as intuitive cues to reconcile competing attentional demands in mobile AR.

Index Terms: Augmented reality, context-aware interaction, adaptive interface, dual task.

1 INTRODUCTION

As Augmented Reality (AR) Head-Mounted Displays (HMD) become increasingly portable, users frequently interact with digital content while navigating physical environments [7, 5, 6]. However, this continuous interaction divides limited attentional resources, introducing substantial safety risks such as collisions or reduced situational awareness [1].

While prior research has explored the effects of elements [3] and context-aware adaptive interfaces [4, 2], these approaches typically target stationary users or discrete location changes, neglecting the continuous environmental dynamics of locomotion. For obstacle avoidance, current solutions predominantly rely on explicit visual feedback paradigms, such as warning overlays or auditory alarms [1]. While effective, these explicit visual cues often exacerbate visual clutter. In this work, we propose a different paradigm: *Implicit Environmental Cues*. Instead of adding new visual elements to warn the user, we modify the spatial behavior of the existing content.

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We introduce DodgeUI, an adaptive interface that leverages Artificial Potential Field (APF) to dynamically reposition a virtual window in response to real-time obstacle detection. The window dodges physical obstacles, treating the content's movement as a signal of environmental hazards. We hypothesize that this behavior maintains interaction continuity while providing a subtle, non-intrusive signal that guides user attention toward safety risks.

2 SYSTEM AND METHOD

The system comprises two main components: a sensing module that captures environmental context and a planning module that determines the adaptive behavior of the user interface.

2.1 Visual Sensing and Obstacle Detection

The sensing module captures RGB-D data from a Microsoft HoloLens 2 headset. To prioritize semantic robustness and detection accuracy for experimental validation, we offload processing to a remote server running Grounding-SAM2 [8]. This ensures the system can identify diverse open-world obstacles with rich semantics, providing a reliable input stream for the motion planner. We then apply depth-based filtering to generate a binary obstacle mask, isolating only relevant physical objects within a 4.0m influence radius to focus strictly on immediate navigational hazards.

2.2 Adaptive Positioning via Artificial Potential Field

We employ the APF algorithm for planning to translate this obstacle data into smooth interface movement. Unlike predictive path-planning methods, this reactive framework minimizes computational overhead, ensuring real-time responsiveness to dynamic environmental changes on resource-constrained mobile hardware. Focusing on a single active interaction window in AR, we treat the window as a virtual particle in the dynamic force field defined by two competing vectors: **Attractive Force** (Anchor): A virtual spring pulls the window toward a pre-defined anchored position, ensuring the interface returns to visibility when the path is clear; **Repulsive Force** (Obstacle): Detected pixels in the obstacle mask exert a repulsive force inversely proportional to their distance.

This physics-based approach creates a real-time continuous movement as implicit cues and adapts to various situations. Distant obstacles generate weak repulsive forces, causing the window to drift slightly—a subtle nudge indicating a presence without breaking focus. As obstacles approach, repulsive forces increase, pushing the window significantly to clear the user's view. In critical col-

lision scenarios, the repulsive force overcomes the anchor entirely, pushing the window off-screen utilizing the limited field of view of HMD. This disappearance acts as an emergency signal, forcing the user to engage with the physical world.

3 USER STUDY

We conducted a $3 \times 3 \times 2$ within-subject user study to evaluate the efficacy of implicit cues. Participants navigated a 12m physical path while performing a Continuous Performance Task — responding to specific numbers on the AR display. We compared three user interface conditions: *Fixed UI*, anchored relative to head orientation as baseline; *Alert UI*, flashing red peripheral border upon obstacle detection; *DodgeUI*, adaptive positioning. Trials included varying obstacle types (static furniture or dynamic pedestrians) and proximities (Far, Near, Close). We employed a Balanced Latin Square design to counterbalance the presentation order of the interfaces.

During the experiment, we logged objective metrics including walking speed, path deviation, eye gaze, and task accuracy. Subjective measures (NASA-TLX, SUS) and qualitative interviews were collected after each block.

4 RESULTS AND DISCUSSION

A total of 24 participants (14 male, 10 female; aged ($M = 22.8, SD = 3.5$) were recruited from local university community.

Objective Safety & Attention. Analysis of walking speed revealed a significant main effect of UI ($F_{2,46} = 6.37, p < .005$), with DodgeUI ($M = 0.79, SD = 0.08$) indicating a slightly lower speed than Fixed UI ($M = 0.81, SD = 0.08$), ($p < .05$). Critically, we found no significant main effects on path deviation ($\chi^2(2) = 0.58, p > .05$), interaction accuracy ($F_{2,46} = 2.84, p = .059$), and response time ($F_{2,46} = 0.75, p = .47$). Qualitative feedback confirmed that visual motion prompted switching from task to observing surroundings, leading to reduced speed. This combined pattern indicates that users effectively traded velocity for increased context awareness without sacrificing dual-task stability.

Furthermore, eye-tracking data substantiates the mechanism of implicit attention guidance (see Fig. 2). We observed a significant reduction in window gazing time for DodgeUI compared to the other interfaces. This demonstrates that the window’s movement successfully prompted a shift in visual focus from the virtual content to the physical environment during potential hazards, verifying the intended signaling effect.

Subjective Preference & Insight. Subjective assessments of workload (NASA-TLX) and usability (SUS) remained consistent across conditions, suggesting that the adaptive motion did not introduce additional task load. However, despite this parity in perceived effort, users’ preferences shifted in high-risk scenarios (see Fig. 3). As obstacle difficulty increased, user preference shifted significantly toward the adaptive interface. This indicates that users recognize the value of the implicit cue in mediating safety. Qualitative interviews also supported our *Implicit Cue* hypothesis, with participants noting, [I] feel like following the window as it goes, even sometimes I haven’t noticed too much about the obstacles.

5 CONCLUSION

In this work, we introduced DodgeUI, an adaptive AR interface that leverages the intrinsic motion of virtual content as an implicit environmental cue. By transforming the avoidance behavior of the window into a communicative signal, we enable users to maintain context awareness in dynamic, high-risk scenarios without increasing visual clutter. The results from the user study show minimal objective performance difference but a subjective preference for the adaptive interface in more complex conditions. Moving forward, we aim to implement this interaction paradigm by optimizing computation efficiency for standalone devices and exploring hybrid models that blend implicit motion with explicit visualizations.

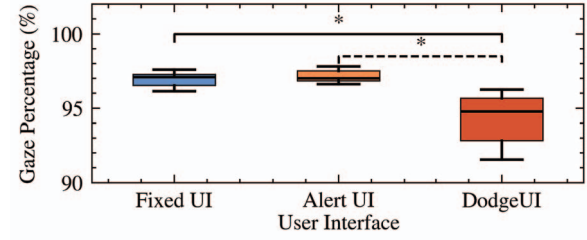


Figure 2: **Percentage of eye gaze duration on virtual window.** DodgeUI significantly reduced visual fixation on the content compared to baselines (* $p < .05$).

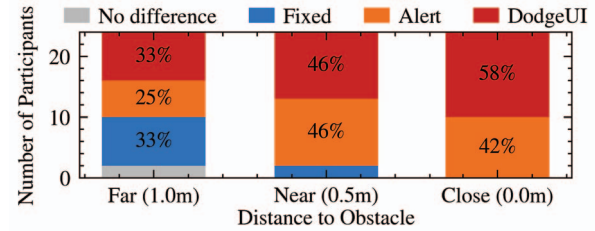


Figure 3: **User preference for UI.** Preference shifts significantly toward DodgeUI (Orange) as obstacles become closer with higher risk.

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